higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 2

(12020132)

12 March 2019 (X-Paper)
09:00–12:00

Calculators may be used.

This question paper consists of 6 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NC (V)

QUESTION 1: GENERAL INTRODUCTION TO FORMWORK AND FALSEWORK

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Formwork is used in the construction of concrete columns and is also called shuttering.

1.2 Formwork is used to contain freshly placed concrete that must be compacted, and after compaction the formwork can be removed immediately to be used for the next task.

1.3 Formwork is a temporary or permanent structure or mould to produce a specific concrete element of the required size and shape.

1.4 After fresh concrete has been placed, it takes 28 days to reach its maximum strength.

1.5 Shutters don't have to be erected level because the gravity of the concrete will level itself.

1.6 Forms should be set up with the joints tight enough to prevent leakage of cement paste.

1.7 Formwork constructed of timber does not need releasing agent because concrete does not stick to timber.

1.8 Rubble and dust do not need to be removed from formwork; it saves time and the wet concrete absorbs it anyway.

1.9 Honeycombs in concrete are good for nature and the construction industry.

1.10 Blowholes are small holes of less than 15 mm caused by air trapped between the formwork and the concrete faces.

(10 × 1) **[10]**

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QUESTION 2: FORM FACE MATERIALS AND SPECIAL TYPES OF FORMS

2.1 In the building industry, steel formwork becomes more popular every day.

2.1.1 Name FOUR advantages of using steel for formwork.

2.1.2 Name FOUR disadvantages when using steel formwork.

(4 × 2) (8)

2.2 What is the difference between permeable and non-permeable form face material? (2)

2.3 Flying table form systems consist of slab formwork tables.

2.3.1 For what type of structures is flying table formwork used? (1)

2.3.2 Explain how 'flying table formwork' is moved from one floor to the next. (2)

2.4 Name TWO advantages of permanent formwork. (2)

[15]

QUESTION 3: PRESSURES RESISTED BY FORMWORK AND FALSEWORK

3.1 Calculate the hydrostatic pressure at the bottom of a dam wall if the water level is 8 m above the bottom of the dam. The density of water is 1 000 kg/m^3 and gravity is 9,8 m/s^2. Use the following formula: (P = pgh) (5)

3.2 When placing wet concrete, the pressure increases as the depth of the concrete increases.

Make a neat triangular hydrostatic diagram to explain the hydrostatic pressure at the bottom of the concrete. (4)

3.3 Give a short definition of the term *concrete form ties*. (1)

[10]

QUESTION 4: ERECTING FORMWORK AND FALSEWORK

4.1 Name the parts of a prop as seen in FIGURE 1 below. Write only the answer next to the question number (1.1–1.10) in the ANSWER BOOK.

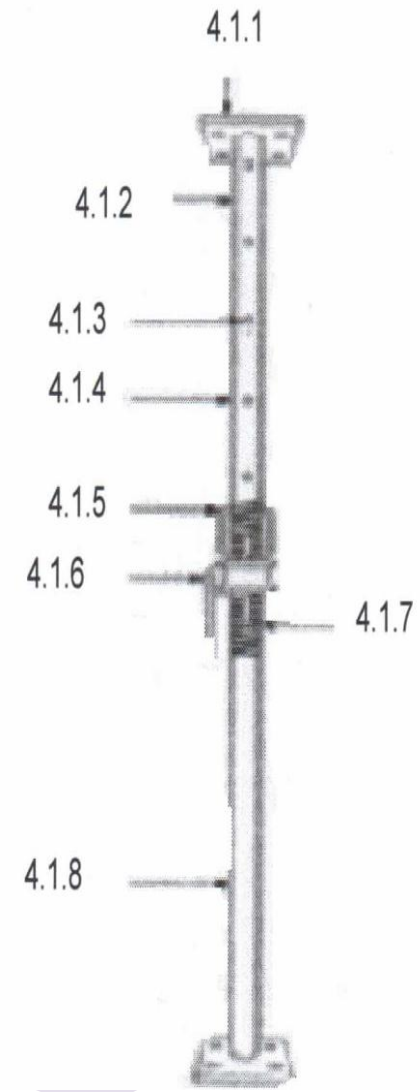


FIGURE 1 (8)

4.2 Draw a neat horizontal section of a formwork for a 600 × 600 mm concrete column made of 20-mm shutter board. Label the drawing and show dimensions.

NOTE: Show formwork only. (11)

4.3 Name FOUR steps that need to be followed when erecting support for concrete slab formwork. (4)

4.4 You need to do two important things before erecting shuttering for formwork slabs and beams.

What are these TWO important things? (2)

[25]

QUESTION 5: VERTICAL STOP ENDS, WATER STOPS AND JOINTS

5.1 Describe the following joints:

5.1.1 Construction joint

5.1.2 Movement joint

5.1.3 Expansion joint

(3 × 2) (6)

5.2 What basic rules can be followed to avoid cold joints? (2)

5.3 Give TWO reasons for delay in the construction process that could lead to a cold joint. (2)

[10]

QUESTION 6: USING RELEASING AGENTS

6.1 Release agents contain substances that can be harmful to your health and that are a safety and fire risk.

Name FOUR safety practices when working with or handling releasing agents. (4)

6.2 What tools or equipment can you use to apply a releasing agent? (4)

6.3 How long after applying a releasing agent should you cast concrete? (1)

6.4 How long does it take concrete to reach its maximum strength? (1)

[10]

QUESTION 7: WORKMANSHIP AND DETAILING IN CONCRETE STRUCTURES

7.1 Draw up an inspection list of SIX of the general items to be checked before concrete is cast. (6)

7.2 By means of a neat drawing, indicate how to reduce grout leakage at a panel corner. (10)

7.3 Make a neat, labelled, freehand sketch to show the effect of rebate on a concrete cover. (4)

[20]

TOTAL: 100

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